

TEACHING PHILOSOPHY

Inquiry with Structure • Rigor with Access • Revision with Purpose

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Students should leave science class able to ask better questions, use evidence, explain systems, and revise their thinking.

WHAT I BELIEVE

Science becomes meaningful when students use it to explain something that matters. I organize learning around phenomena, problems, and systems that create a genuine need for new ideas. Content knowledge is essential, but students should do more than recognize vocabulary: they should model relationships, interpret evidence, challenge explanations, and apply what they know in unfamiliar situations.

INQUIRY WITH STRUCTURE

Inquiry is not vague discovery. I use concise instruction, strong models, purposeful questions, visual supports, and checkpoints so students can investigate with increasing independence.

RIGOR WITH ACCESS

My special education experience taught me that high expectations require multiple entry points, predictable routines, calm relationships, and flexible ways for students to show understanding.

ASSESSMENT FOR REVISION

Models, lab evidence, discussion, CER writing, conferences, and quick checks help me identify misconceptions early. Feedback is actionable, and revision is an expected part of scientific work.

WHAT THIS LOOKS LIKE IN PRACTICE

In a photosynthesis and cellular respiration sequence, I might begin with a visible change in a BTB-Elodea investigation, ask students to propose explanations, and then use modeling, targeted instruction, and discussion to trace matter and energy through the system. Students compare their initial ideas with evidence, write a claim-evidence-reasoning explanation, and revise the model when the evidence no longer fits. The lesson is active, but every activity serves a clear scientific purpose.

LEARNING ACROSS CULTURES AND DIFFERENCES

A strong classroom treats language, culture, and perspective as resources. Having taught international students and supported neurodiverse learners, I aim to build teams where students listen carefully, disagree productively, and contribute in meaningful ways. Technology is used when it improves access, makes a system visible, enables testing, or gives students a stronger way to communicate—not simply because it is available.

MY ROLE AS THE TEACHER

My role is to make complex ideas accessible without removing the intellectual work. I listen closely to student reasoning, decide when to question, model, explain, or step back, and create enough structure that students can take genuine ownership of the next step.

My goal is an intellectually ambitious, emotionally safe classroom where students investigate before accepting, explain rather than guess, and improve their work through evidence and reflection.